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Is Paraconsistent Logic the Logic of Mathematics?

Bachelor Thesis

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Abstract

In classical logic, any statement can be proven from a contradiction. In paraconsistent logic, this is not the case. This thesis examines some of the methods to formalize paraconsistent logics, in particular for the logic of paradox as well as Chunc and Permeate. The thesis also connects mathematics to paraconsistency and demonstrates some applications. From a philosophical point of view we inspect the motivations for paraconsistency, what logic is used by mathematicians and if inconsistencies exist.

Contents

Contents	iii
1 Introduction	1
1.1 Classical Logic	1
1.1.1 Principle of Explosion (ECQ)	4
1.2 Inconsistencies and Paradoxes	4
1.3 Paraconsistency	5
1.4 Examples	7
2 Formalization of Paraconsistency	9
2.1 LP	9
2.2 Challenges	11
2.3 Formalization of other Paraconsistent Systems	14
2.4 Reflection	15
3 Inconsistent Systems of Mathematics and Paraconsistent Logic	17
3.1 Inconsistent Models of Arithmetic	17
3.2 Chunk and Permeate	19
3.2.1 Dirac's Delta Function	20
4 Philosophy of Paraconsistency	25
4.1 Motivations for Paraconsistency	25
4.2 What Logic is used by Mathematicians?	27
4.3 Do Inconsistencies Exist?	29
5 Conclusion	31
A Appendix	33
A.1 Deduction via Tarski Truth Scheme	33
Bibliography	35

Chapter 1

Introduction

In this work, we will take a look at paraconsistent logic and how it relates to mathematics. To do this, I first want to give a gentle introduction to the structures that are used for reasoning in logic and the problems that come with contradictions.

1.1 Classical Logic

In this section I will describe classical logic. The main idea to take away is given in the subsection 1.1.1 about the Principle of Explosion (ECQ).

Classical logic is a modern description of how to formalize logic. The word classical does not refer to ancient Greece or any classical period. Classical or standard logic was developed in the late 19th and early 20th century. Some call it a mathematical model for reasoning or language itself [SKK21]. The basis of first order classical logic is an alphabet which is a set of symbols also called letters. Some common example is:

Category	Symbols	Usage in a sentence
Constants	$0, 1, a, a_1, a_2, \dots$	standalone
Variables	$x, x_0, \dots, y, y_4, \dots$	standalone
Order	$() , []$	used for order
Predicate	$=, P, Q, R, >, \leq$	$x \leq a$ or $Q(x, y, z)$
Function	$+, -, f, f_1, \dots, g,$	$1 + x$ or $f(x, y, a)$
Connective	$\neg, \wedge, \vee, \rightarrow, \leftrightarrow$	$[g(x - a) \leq b] \rightarrow [g(f(x - a)) \leq c]$
Quantifier	$\forall, \exists$	$\forall x([g(x - a) \leq b] \rightarrow [g(f(x) - f(a)) \leq c])$

Constants model references to specific values. At this point it is not yet clear what these values are. For example we can have the constant π . It just references to a fixed value, which is not yet determined. Variables are used

1. INTRODUCTION

to model generality and can be connected to quantifiers¹ which we will not focus on in this work.

The predicates map constants, variables or functions to either true or false. A n -place predicate with n variables or constants is called an atomic formula. These atomic formulas can then be chained together with connectives, also called logical operators, to formulate complex well formulated formulas. The negation ($\neg$) is placed in front of a formula. Where as the and ($\wedge$), or ($\vee$), implication ($\rightarrow$), equivalence ($\leftrightarrow$) connect exactly two formulas. We will take a quick look at how these operators work, when applied to so called closed formulas or sentences. The result of the connection can be shown in truth tables. Here, t and f stand for true and false respectively. The truth value in the row names stands on the right side of the operator.

$\neg$		$\wedge$	t	f	$\vee$	t	f
t	f	t	t	f	t	t	t
f	t	f	f	f	f	t	f

In a more numerical description of the operators², we first associate *true* with 1 and *false* with 0. Let A and B be either true 1 or false 0 then we get the formulas:

$$\begin{aligned}\neg A &= 1 - A \\ A \wedge B &= \min(A, B) \\ A \vee B &= \max(A, B)\end{aligned}$$

For any formulas A or B . The connections made by the implication operator $A \rightarrow B$ is equal to $A \vee \neg B$ and the equivalence operator $A \leftrightarrow B$ to $(A \rightarrow B) \wedge (B \rightarrow A)$. Quantifiers are used to bind variables. See [SKK21] for a detailed description.

A sentence is a finite sequence of letters from our alphabet. If the syntactical rules we quickly touched on are fulfilled then we call the sentence well-formed. I will write sentence instead of well-formed sentence in the rest of this work³. At some point, we will have a set of sentences, say Γ , that are all true. From these we want to deduce new sentences.

An example: Let the sentences A and $\neg A \vee B$ both be true. Because A is true $\neg A$ is false. So $\neg A \vee B$ can only be true if B is true⁴. To make such arguments reusable, we define inference rules. These have some premisses

¹A formula is called closed if all variables are connected to quantifiers, otherwise it is called open.

²Based on Boolean algebra.

³Because we will always talk about well-formed ones.

⁴Check the truth table of $\vee$ above to convince yourself.

and a conclusion. An inference rules is said to be sound, if we can show that if the premisses are true, then the conclusion is also true. Here are two inference rules that we will also consider later on. For more rules and argumentation for their soundness see [SKK21].

$$\frac{A \quad \neg A \vee B}{B} \text{(disjunctive syllogism)} \quad \frac{A \quad A \rightarrow B}{B} \text{(modus ponens)}$$

The premisses are above the horizontal line and the conclusions below. As you can see they are very simple. If we chain multiple inference rules together we get a deduction, also called formal proof. The notation for a deduction is $\{A, B, \dots\} \vdash Z$ where A, B, Z are sentences. Meaning that from the set of sentences $\Gamma := \{A, B, \dots\}$ we can deduce Z by inference rules⁵.

Now we will focus on the semantic, which gives meaning to sentences. Here I want to go over two different logic systems. In propositional logic⁶ we only consider connectives and 0-valued predicates, also called sentences letters $P := \{Q, R, S, \dots\}$. The interpretation function $\nu : P \rightarrow \{t, f\}$ defines which of these sentence letters are true and which are false. Let L be the set of all sentences we can build by connecting sentence letters with connectives. We can extend ν to $\nu^+ : L \rightarrow \{t, f\}$ by using the truth tables describes above. Thus ν^+ can evaluate the truth value of any sentence in L .

In first-order logic we have more letters in our alphabet. An interpretation or model $M = \langle I, D \rangle$ consists of a non-empty set D called domain of discourse and an interpretation function I . The domain of discourse are the objects we want to talk about. Mathematical examples could be the set of all natural numbers $\mathbb{N} := \{0, 1, 2, 3, \dots\}$. But we could also consider all the people who ever existed or all the things that exist in the universe. The interpretation function gives meaning to the constants, predicates and functions. For some constant c in the alphabet the interpretation is in the domain of discourse $I(c) \in d$. For an n -valued function $f = f(x_1, \dots, x_n)$ the interpretation is $I(f) : D^n \rightarrow D$ a function over D .

The consequence relation in first order logic $M \models A$ is given if A is interpreted as true under the model M . If A has no free variables⁷ then this is similar to propositional logic. If A contains free variables then A is true when it is true for all variable assignments of the free variables in A to any element in the domain of discourse.

As an example in first-order logic we consider the sentence A containing the letters $x \geq 0$, where x is a variable $\geq$ a predicate and 0 a constant. First we use a specific interpretation function I_0 that makes 0 and $\geq$ behave as we are

⁵If we take some initial set Γ and claim its sentences to be true then the sentences in Γ are called axioms.

⁶Also called zeroth-order logic, because it only used 0-valued predicates and connectives.

⁷Non-free variables are the ones that are connected to quantifiers.

used to in mathematics. If we take $\mathbb{N}$ as the domain of discourse, then A is true for all assignments for variable x in the model $M_1 := \langle \mathbb{N}, I_0 \rangle$ and thus $M_1 \models A$. But if we take $\mathbb{Z} := \{\dots, -1, 0, 1, \dots\}$ as the domain of discourse, then A is false in the model $M_2 := \langle \mathbb{Z}, I_0 \rangle$ because if we assign -2 to x then $x \geq 0$ is false. Therefore $M_2 \not\models A$, where $\not\models$ is just *not* $\models$ analogous to $\neq$.

We say that an argument $\langle \Gamma, A \rangle$ is valid, written $\Gamma \models A$, if for every model M and all sentences B in Γ we have $M \models B$ and also $M \models A$. The soundness theorem states:

Theorem 1.1 *For any sentence A and set of sentences Γ , if $\Gamma \vdash A$ then $\Gamma \models A$. Proof in [SKK21].*

1.1.1 Principle of Explosion (ECQ)

One property of classical logic is called *ex contradictione sequitur quodlibet* (ECQ), meaning *from contradiction anything follows*. This is also named principle of explosion because it leads to the explosion of truth. This is a fundamental aspect of this thesis.

The principle of explosion states that if the sentences A and $\neg A$ are both true, then we can infer any other sentence B .

Proof Let the sentences A and $\neg A$ be true. We take any other sentence B . Then the sentence $A \vee B$ is true, because A is true⁸.

In the next step we consider that $\neg A$ is true and $A \vee B$ is true. Here we use the inference rule described above (disjunctive syllogism) and derive B . $\square$

In a schematic representation:

$$\frac{\neg A \quad \frac{A}{A \vee B} \text{ (disjunction introduction)}}{B} \text{ (disjunctive syllogism)}$$

In this schematic deduction, multiple inference rules are chained together. Note that the starting points are A and $\neg A$. So the premisses are $A, \neg A$ and the conclusion is B . For B we can take any sentence. Therefore we can deduce any sentence B from an inconsistency.

1.2 Inconsistencies and Paradoxes

Formally a set of sentences Γ is consistent if there is no sentence A such that $\Gamma \vdash A$ and $\Gamma \vdash \neg A$. A set of sentences Γ is called inconsistent if $\Gamma \vdash A, \neg A$. The notation here is that both A and $\neg A$ can be deduced from Γ .

⁸See truth tables.

In spoken language inconsistencies appear in form of paradoxes. One example is the liar paradox:

This sentence is a lie.

Let us call this sentence L . If we say that L is true, then L is a lie, which means L is not true. So L implies $\neg L$. Vice versa, if we say that L is false, then L is not a lie and thus true. So $\neg L$ implies L . The problem we face is that we claim that L has to be either true or false but it is somehow $L \leftrightarrow \neg L$.

A similar and famous such paradox arises in mathematics, more specifically, in naive set theory. The prefix naive is exactly because of the existence of this example. In this theory, there are sets that contain elements and there is no restriction on what these elements can be. Importantly, sets can contain sets as elements. The paradox we focus on is called Russell's Paradox. To demonstrate this, we consider the Russell set R , which contains all sets that do not contain themselves.

$$R := \{x | x \notin x\}$$

The question we now ask is if R contains itself. For this we formulate the sentence

$$A := "R \in R"$$

. If A is true, then R is contained in R and thus satisfies the property that $R \notin R$, which means $\neg A$ is true. And vice versa, if A is false, then $R \notin R$ and so R , satisfies the conditions for the elements in R and $R \in R$. Therefore A is true. We get the equivalence:

$$R \in R \leftrightarrow R \notin R$$

This is nothing else than

$$A \leftrightarrow \neg A$$

which is an inconsistency, and thus not acceptable in a theory where sentences are either true or false. Such systems are also called bivalent because there only exist two truth values. Thus, from a classical perspective, such a Russell set R cannot exist. To prevent these kinds of paradoxes, other set theories were developed. The most important ones are the Zermelo-Frankel set theories ZF and ZFC, which can be used as a foundation of most modern mathematics [Bag20]. But these new axiomatic theories cannot guarantee a consistent system either.

1.3 Paraconsistency

Logical systems that have ECQ and inconsistencies at the same time become trivial, meaning that all sentences become true as shown in 1.1. Intuitively

this might seem wrong. Let us consider the naive set theory with the Russell's paradox. With ECQ we can show that "all sets have to have exactly three elements". Clearly, the aim of the naive set theory is not that such a sentence is true. We want to prevent this. One way to do that is to revisit the premisses which lead to contradictions and remove or change some of them.

Another approach is to use a paraconsistent logic. A logical system that has inconsistencies but does not become trivial is called paraconsistent. And this is what we will explore in this work. Paraconsistent logical systems do not follow the ECQ principle and thus are non-explosive. This is because we do not want our theory to become trivial but allow inconsistencies at the same time. Paraconsistency is also strongly connected to a philosophical stance called dialetheism. Dialetheists believe in true contradictions. They think that contradictions such as the Liar paradox *exist* and are *both* true and false at the same time. This is an opposing view to bivalence. It is important to note that dialetheists do not claim all contradictions to be true.

According to Aberdein and Read, paraconsistent research is split into two programs [AR09]. The weakly paraconsistent program focuses on inconsistent information. This might occur in a corrupt database, in conflicting laws or confusion in the development of science. Some of the sentences may be more true than others, so we can discriminate between them. This makes their inference system paraconsistent. The stronger paraconsistent research program is connected to dialetheism. This is motivated by the belief that the world or mathematics are in fact inconsistent. There is a subscript of stronger paraconsistency that does not require a strict belief in the inconsistency of the world, but only agnosticism about it. One of the differences between the two research programs is that weakly paraconsistent systems differ only by logical proof or inference system from the classical approach, while stronger paraconsistency and dialetheism demand to diverge from the classical idea of bivalence.

In this work we will take a closer look at two paraconsistent systems. The logic of paradox 2.1 is a system in the sense of the strong research program because it explicitly allows contradictions to exist. While Chucn and Permeate 3.2 is more a weakly paraconsistent system. It tries to contain consistent systems and permeate information between them.

The word paraconsistent goes back to Francisco Miró Quesada Cantuarias and the Third Latin America Conference on Mathematical Logic in 1976 [PTW18]. The prefix *para* has multiple meanings in English such as against, near or beyond. It is not certain what the original idea was. Some version of paraconsistency can be better described by near consistency, while others are against or even beyond consistency.

1.4 Examples

Before we dive deeper into paraconsistency, I want to quickly go over some examples of systems that appear to be inconsistent. To describe such systems a paraconsistent logic is needed, otherwise the contradictions would make the theory trivial by ECQ and the existence of inconsistencies.

Some very practical examples lie in denotic logic which is concerned with obligation and permissions. In morality or law, there are situations with inconsistent principles or laws. In physics, Bohr's description of the atom was inconsistent [BP15]. It is still used to teach subatomic principles on some education levels. At its time, it helped to develop quantum mechanics. Another example from physics is the Dirac delta function, which is zero at all points besides the origin, but its integral over the whole domain has to evaluate to 1. I will go deeper into this topic in 3.2.1.

In mathematics, the theory of derivatives and integrals which is based on infinitesimal numbers by Leibniz and Newton is inconsistent [Bro02]. This is because infinitesimals are sometimes bigger than zero while at other times equal to zero. This is not how we define derivatives and integrals today, but infinitesimals are still used in teaching physics for example. Another mathematical theory we have already touched on is naive set theory, which describes the intuitive understanding of sets most people use when thinking about or working with sets.

Inconsistencies can also be found in art. For instance, Reutersvård and Escher are famous for producing inconsistent artworks. Mortensen gives a description of some elements in inconsistent drawings [Mor10] and gives a logical argument for their inconsistency. He also connects the inconsistencies to cognition and mathematics.

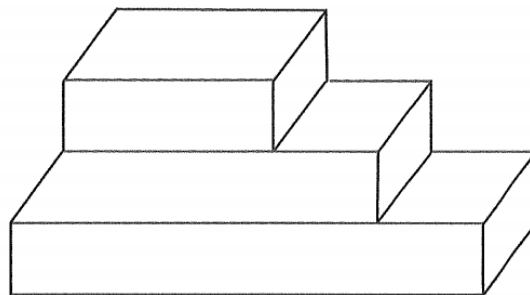


Figure 1.1: Stairs from [Mor10] on page 129.

Chapter 2

Formalization of Paraconsistency

There are a of different formalizations for paraconsistent systems. Overviews and comparisons have been done [Bro02]. I will introduce one such system called Logic of Paradox (LP). Graham Priest is one of the main proponents of LP [Pri79]. We look at LP because it has a simple approach, is truly paraconsistent and can show the challenges of formalization. LP explicitly allows contradictions, thus it aligns itself with the ideas of dialetheists, like Priest himself.

2.1 LP

The main idea of LP is to introduce a new truth value. This approach is inspired by Kleene's three-valued logic [Kle52]. We depart from the bivalence point of view. A sentence that is both true and false is called 'paradoxical' (p). While a sentence that is true but not false is called 'only true' (t) and if a sentence is false but not true it is called 'only false' (f). Let the numeric interpretation of paradoxical be 0.5 in between 0 and 1.

Truth value	numerical value
t	1
p	0.5
f	0

Similar to classical logic, we can define truth tables for different logical operators. To argue for the correctness of these truth tables we extend the notion of the numerical meaning of the logical operators to the paradoxical value 0.5. The negation of only true and only false sentences behave as they do in classical logic. The logical negation $\neg$ in LP can be seen as the extension of $\neg$ from classical logic. The numerical negation of paradoxical is

2. FORMALIZATION OF PARAConsISTENCY

$$-0.5 = 1 - 0.5 = 0.5.$$

$\neg$	
t	f
p	p
f	t

In a similar fashion we can develop truth tables of the conjunction $\wedge$ and disjunction $\vee$. The numerical analog to $\wedge$ is the *min* function and for the $\vee$ the *max* function for the classical values $1, -1$. Here we just extend this thinking to the paradoxical value.

$\wedge$	t	p	f
t	t	p	f
p	p	p	f
f	f	f	f

$\vee$	t	p	f
t	t	t	t
p	t	p	p
f	t	p	f

For the implication and equivalence we can use the previous tables with the identification of the implication $A \rightarrow B$ with $\neg A \vee B$ and of the equivalence $A \leftrightarrow B$ with $(A \rightarrow B) \wedge (B \rightarrow A)$.

$\rightarrow$	t	p	f
t	t	p	f
p	t	p	p
f	t	t	t

$\leftrightarrow$	t	p	f
t	t	p	f
p	p	p	p
f	f	p	t

With this we have defined the elementary logic operators in a three-valued logic. Here, Priest notes that in LP both t and p are designated values, meaning they denote true sentences. This is in contrast to other multivariate logic systems.

Now we want to see how this compares to classical logic. For this, we define an evaluation function $\nu : P \rightarrow \{t, f, p\}$ ¹ that states if a sentence letter is true, false or paradoxical. P is the set of all sentence letters. With them and the elementary operators defined above, we can now build sentences. Let L be the set of all sentences we can build and $\nu^+ : L \rightarrow \{t, f, p\}$ be the natural extension of ν to these sentences with the above truth tables. So if we have a sentence A where some sentence letters are combined with the elementary logic operators in a syntactically correct way, ν^+ can evaluate A .

In LP for a subset of sentences Γ of L we define the logical consequence relation: $\Gamma \models A$ iff there is no ν such that $\nu^+(A) = f$ but for all $B \in \Gamma$ we have $\nu^+(B) = t$ or p . Meaning that A follows from the sentences Γ iff A is not evaluated as false by ν^+ while all sentences $B \in \Gamma$ are evaluated as either true or paradoxical.

¹In the numerical sense $\nu : P \rightarrow \{0, 0.5, 1\}$.

Tautologies are sentences that are true no matter how you interpret their sentence letters. They are written as $\models A$. Before the $\models$ we do not have a set of sentences, or in other words, we have an empty set of sentences $\emptyset \models A$. So for all interpretations ν , $\nu^+(A) = t$ or p . Now, we check if tautologies in classical logic are consistent with tautologies in LP.

Theorem 2.1 *A is a two-valued tautology if and only if A is a three-valued tautology $\models A$.*

Proof " $\leftarrow$ " from right to left: If A is a three-valued tautology, then it is also a two-valued tautology. If we ignore the paradoxical truth values, A is still a tautology with the classical truth values.

" $\rightarrow$ " from left to right: We take ν , a three-valued evaluation. And ν_2 a two-valued extension of ν , where we map p evaluations of ν also to t . By the truth tables above it follows that if $\nu^+(A) = f$, then $\nu_2^+(A) = f$.

Now we argue by contradiction. Given a two-valued tautology A , meaning for all two-valued evaluations we get $\nu_2^+(A) = t$. If $\nu^+(A) = f$ then $\nu_2^+(A) = f$ which is a contradiction to our assumption of A . Thus there exists no ν such that $\nu^+(A) = f$ and thereby A is per definition a tautology. $\square$

This theorem is a first step to preserving logical intuition of classical logic in LP. The tautologies in LP are the same as the tautologies in classical logic, because $\models A$ means $\nu^+(A) = t$ or p for all ν^+ .

Theorem 2.2 *If A and B have no sentence letters in common and if B can take the value f, then $A \not\models B$.*

Proof Let ν be an evaluation such that $\nu^+(B) = f$. Let $\bar{\nu}$ be like ν but such that for all sentence letters Q in A , the evaluation is paradoxical, $\bar{\nu}(Q) = p$. We can assume this because these k do not occur in B , thus we have no other assumptions about them. By the above truth tables we get $\bar{\nu}^+(A) = p$. Note that if we connect paradoxical values with any of the logical operators the result is paradoxical.

By the definition of $\models$ given above, B does not follow from A . $\square$

2.2 Challenges

A lot of the inferences that we use in mathematics are valid in LP. But some crucial ones are not. Examples are 'disjunctive syllogism', 'modus ponens' and 'modus tollens'. The disjunctive syllogism is the rule that if sentences A and $\neg A \vee B$ are both true, you can deduce B . Modus ponens states: If you have A and $A \rightarrow B$ then we can infer that B is true. Modus tollens states: If

you have $A \rightarrow \neg B$ and B then you can infer $\neg A$.

$$\frac{A \quad (\neg A \vee B)}{B} \text{ (disjunctive syllogism)} \quad \frac{A \rightarrow B \quad \neg B}{\neg A} \text{ (modus tollens)}$$

With the classical truth values t and f , the mentioned rules are sound in classical logic [SKK21] and so they are sound in LP when only t and f are present. But consider modus ponens

$$\frac{A \quad A \rightarrow B}{B} \text{ (modus ponens)}$$

Let A be p , so both true and false. Let B be false, then A and $A \rightarrow B$ can both be evaluated as true, but B is not. Thus ‘modus ponens’ is a problematic rule in LP. These rules are value-preserving but not truth preserving.

Priest uses the name quasi-valid for problematic rules. For classical truth values everything behaves accordingly but for p some inference rules become invalid. We want a system which has all or most of the capabilities of classical logic but which can handle inconsistencies. There are two suggested ways of dealing with the issue of quasi-valid rules.

In the first approach we remove problematic inference rules. We first consider a language subset of LP which only contains the logical connections $\neg$, $\wedge$ and $\vee$. Here we take a look at the disjunctive syllogism described above. This is exactly what the principle of explosion (EFQ) describes. If A is p , then A and $\neg A$ are both true. Because A is true, also $A \vee B$ is true. Now we apply the disjunctive syllogism and get that B is true. So this rule has to go.

Next, we look at the implication $\rightarrow$. Some problematic principles are modus ponens, modus tollens and reductio ad absurdum. For the meaning of $A \rightarrow B$ we used $\neg A \vee B$. This is a variant of the disjunctive syllogism. So maybe the cause of our problem is the definition of $\rightarrow$. Here, we need to look at the following two rules:

$$\frac{A \quad A \rightarrow B}{B} \text{ (modus ponens)} \quad \frac{A \rightarrow (A \rightarrow B)}{A \rightarrow B} \text{ (absorption)} ..$$

If $\rightarrow$ fulfills these, then we can deduce anything. Consider the sentence ‘If this sentence is true, then A is true’. Formulated symbolically:

$$(1) \quad T(1) \rightarrow A.$$

Here Tx stands for x is true. The (1) is the name of the sentence and is used as a self-reference. With the Traski truth scheme² we get:

$$(2) \quad T(1) \leftrightarrow (T(1) \rightarrow A).$$

²See [Hod18] for more explanation.

For the implication ' $\rightarrow$ ' in the equality $\leftrightarrow$ in (2) we use absorption and get

$$(3) \quad T(1) \rightarrow A$$

and with modus ponens for ' $\leftarrow$ ' in (2) and (3) we get

$$(4) \quad T(1).$$

Finally by using modus ponens, with premisses (3) and (4) we deduce

$$(5) \quad A.$$

For a schematic explanation of this deduction, see A.1 in the appendix.

This shows that either modus ponens or absorption must be given up. Otherwise, any theory would be trivial. Priest notes that much more work is needed to fully formalize this. When giving up inference rules, the logic becomes weaker and is called substructural logic [Res18]. But a weaker logic is not our goal. We rather want an extension to describe more than classical logic can.

Priest also suggests a different, more radical approach to the problem of quasi-valid inference rules. Giving up an inference rule like modus ponens would cripple reasoning tremendously. He asks:

Why should we not go on using them anyway? (Priest, *Logic of Paradox* [Pri79], p235)

There is no harm in using quasi-valid inference rules if all statements take classical truth values.

The problem now shifts to identifying paradoxical sentences. This is not an easy task because, paradoxical sentences are not marked and may be hidden. Paradoxical sentences often appear in context of self-reference, thus there is hope that they only occur in these circumstances [Pri79] and that only a limited number of paradoxical sentences exist. We do not deal with paradoxical sentences on a daily basis.

Priest formulates a methodological maxim (MM) as a rule of conduct.

MM : Unless we have specific ground for believing that paradoxical sentences are occurring in our argument, we can allow ourselves to use both valid and quasi-valid inferences.

This is a strong argument in view of the status quo and classical logic. But now we cannot be sure of our conclusions, because we have quasi-valid inferences. Priest argues that even if you are certain of your inference rules, your certainty of the conclusion is always based on your certainty of the premisses. If our premisses all behave in a bivalent fashion there is nothing

to worry about. As soon as we detect a paradox, our set of inference rules gets immensely restricted.

This has some consequences on the epistemology in a logical system. Assume that we have an argument for some sentence we believe in. If we believe there are no paradoxical sentences in the proof, then we can invoke MM and claim the sentence to be true. We have reason to claim a sentence to be paradoxical, if we can show it to be true and false. When we later find evidence that some sentence in the argument is paradoxical, we have to revoke MM and check if quasi-valid inferences were used in context with this paradoxical sentence. Is that the case, we need to reject our previous proof. This is nothing new in mathematics and similar to a proof that was believed to be valid and later discovered to be incomplete or non-valid.

2.3 Formalization of other Paraconsistent Systems

In their article, Aberdein and Read discuss the formalization of different paraconsistent systems [AR09]. The first step to formalization is to reject ECQ. A simple approach is to discard the simple rule of adjunction

$$\frac{A \quad B}{A \wedge B} \text{ (rule of adjunction)}$$

because then from $A, \neg A$ we cannot infer $A \wedge \neg A$. Such systems are called non-adjunctive and allow inconsistencies. But this prevents the existence of true contradictions. Therefore, non-adjunctive systems are unsuitable for dialetheists who claim some contradictions to be true. Non-adjunctive systems are also not suited for the weakly paraconsistent program, because adjunction is such a fundamental part of logic. It is hard to see how any non-adjunctive system could represent the conjunction ' $\wedge$ ' which is a primal logical operator.

A different approach is to take a look at the rules for implications. One problematic property is the negative paradox of material implication (NPI) $\neg A \vdash A \rightarrow B$. This leads to ECQ in the presence of modus ponens.

$$\frac{A \quad \frac{\neg A}{A \rightarrow B} \text{ (NPI)}}{B} \text{ (modus ponens)}$$

Paraconsistent systems such as non-adjunctive or broadly relevant ones also drop the positive paradox of material implication (PPI) $B \vdash A \rightarrow B$.

Though it is possible to formulate systems that keep PPI like the systems C_n developed by da Costa [dC74]. These are called positive-plus systems. Keeping PPI results in the sacrifice of modus tollens and transposition ($A \rightarrow$

$B) \vdash (\neg B \rightarrow \neg A)$ [PRN93]. If they are not sacrificed, these systems explode.

$$\frac{\frac{A}{B \rightarrow A} \text{ (PPI)}}{\frac{\neg A \rightarrow \neg B}{\neg B} \text{ (modus ponens)}} \text{ (transposition)}$$

Giving up NPI and PPI limits the potential for paraconsistent systems to recapture classical logic.

2.4 Reflection

Priest concludes his article [Pri79] by saying that what he has written is false. In particular if what he said was true, then some things he stated are false. This is a self reference and play on contradictions themselves. His statement is easy to illustrate. Consider the sentence ' A is only true' where A is any sentence. In LP the sentence A can either be only true ' t ', only false ' f ' or paradoxical ' p ' if it is true and false. Illustrated with a truth table:

A	A is only true
t	t
p	f
f	f

Let us consider an extended Liar paradox.

(6) This sentence is not only true.

If (6) were only true, then (6) is not only true and thus it is false. If (6) is not only false then it is only true. And finally if (6) is paradoxical it is clearly only false. We get that (6) is only true and not only true. This is very similar to the liar paradox and shows that LP is not immune to this kind of reasoning. But that is not the point. Priest does not want to prevent paradoxes but find a way to accept them, because he thinks that they are inevitable.

Paraconsistent logics try to be as powerful as classical logic and at the same time accept inconsistencies and paradoxes, while not rendering every theory trivial by explosion. To do this formally is not simple and comes with many problems as shown in sections 2.2 and 2.3. The main approach seems to be to dismiss fundamental inference rules of classical logic, with the goal to get a system where the principle of explosion does not hold. This clearly limits the deductive power, which is a trade-off that does not seem justifiable to me.

I value the second approach of Priest, where he states that our belief in a sentence depends on the validity of inference rules used in its deduction and

equally on the validity of the premisses. This approach has a simplicity in its execution and does not challenge inference rules. At its core we accept that there are statements that are paradoxical p . However as long as we are not working with them, the rules of classical logic are good enough. As soon as paradoxical sentences appear, we might not want to continue because by ECQ this would turn every theory trivial.

In sections 2.2 and 2.3 we saw the problems of formalizing LP and other paraconsistent systems by cutting away inference rules. I consider it problematic if we want to use paraconsistent logic as the logic of mathematics. There are other interesting applications of such limited systems. But the limitations are too severe to be practical for mathematical reasoning. Most theorems in mathematics are in the form of a simple implication $A \rightarrow B$. To then apply them, we use something analogous to modus ponens. Thus giving up such a basic logical inference rule is very limiting.

Inconsistent Systems of Mathematics and Paraconsistent Logic

Here, we take a look at how a paraconsistent view can be applied to mathematics. First, we inspect inconsistent models of arithmetic [Pri97] as an explanatory exercise. Then, we introduce the Chunks and Permeation technique [BP04] and apply it to the Dirac delta function δ [BMP14].

3.1 Inconsistent Models of Arithmetic

In this section, we use a first-order version of LP. Priest demonstrates in [Pri97] how to create inconsistent models with LP. An LP interpretation is a pair $\langle D, I \rangle$ as is the case in first-order logic. D is the domain of discourse and I the interpretation function for non-logical symbols of the language.

Let M be any model $\langle D, I \rangle$. Let $\sim$ be any equivalence relation on D and a congruence relation on the interpretation of the function symbols [Pri97]. For any n -place function symbol $f(x_1, \dots, x_n)$ if $d_i \sim e_i$ for $d_i, e_i \in D, 1 \leq i \leq n$ then $I(f)(d_1, \dots, d_n) \sim I(f)(e_1, \dots, e_n)$. With $[d]$ we denote the equivalence class for $d \in D$ under $\sim$.

Now we collapse the interpretation to $M^\sim = \langle D^\sim, I^\sim \rangle$ named the collapsed interpretation, where the domain is $D^\sim = \{[d] \mid d \in D\}$. The interpretation $I^\sim$ follows: for every n -place function symbol $I^\sim(f)([d_1], \dots, [d_n]) = [I(f)(d_1, \dots, d_n)]$. The n -place predicate $P = \langle [d_1] \dots [d_n] \rangle$ is in the extension $M^\sim$ iff $e_1 \sim d_1, \dots, e_n \sim d_n$ and $\langle e_1 \dots e_n \rangle$ is an assignment to P in I that is positive.

Now we formulate the collapsing lemma [Pri79] for an evaluation function ν_M which associates f, p, t to every formula:

Lemma 3.1 *For any formula A of L_M we have $\nu_M(A) \subseteq \nu_{M^\sim}(A)$.*

The proof of this can be done via recursion, see [Pri97], [Pri91]. This lemma shows that truth values are never lost by collapsing. A true or false sentence before collapsing is still true or false after collapsing, but may be not only true or only false. If we collapse a model M to $M^\sim$ where $M \models A$, then $M^\sim \models A$.

Now we want to apply this lemma to a model of arithmetic. Let L be the language of arithmetic. There is an equivalence $=$, one constant symbol 0 and function symbols for successor, addition and multiplication $', +, \times$. The number n is represented by 0 followed by n primes $'$. As an example the number 3 is represented as $0'''$.

Let a $M = \langle \mathbb{N}, I \rangle$ be a standard model of arithmetic. The domain of arithmetic is $\mathbb{N}_0 = \{\mathbf{1}, \mathbf{2}, \mathbf{3}, \dots\}$, the bold numbers for elements in the domain of discourse. The interpretation I is the standard arithmetic interpretation. So $I(0''')$ is 3. We have a successor function $'$ which starts off¹ at 0.

$$0 \rightarrow 0' \rightarrow 0'' \rightarrow \dots$$

We can abbreviate 0 with n primes by a non bold n , because we build a finite system.

$$0 \rightarrow 1 \rightarrow 2 \rightarrow \dots$$

In a finite model, at some point two numbers will have identical denotations. Because the domain of discourse is infinite. Let $\mathbf{n}$ and $\mathbf{m}$ be the smallest numbers for which this is the case. Then the successor graph looks as follows:

$$\begin{array}{ccccccc} 0 & \rightarrow & 1 & \rightarrow & \dots & \rightarrow & n & \longrightarrow & n+1 \\ & & & & & & \uparrow & & \downarrow \\ & & & & & & n+m-1 & \leftarrow \dots \leftarrow & n+2 \end{array}$$

If $n = n + m$, then we also have that $(n)' = (n + m)'$ by the successor rule, thus $n + 1 = n + m + 1$. Therefore, we get a loop at that repeats itself indefinitely.

The equivalence relation $\sim$ for $n, m \in \mathbb{N}_0$, is given by

$$a \sim b \quad \text{if } a \equiv b \pmod{m} \text{ and } a, b > n.$$

Now, we collapse the domain to

$$\begin{aligned} \mathbb{N}_0^\sim &= \{[0], [1], \dots, [n], [n+1], \dots, [n+m-1]\}, \\ \text{where } [l] &= \begin{cases} \{k\} & l < n \\ \{l + jm \mid j \in \mathbb{N}_0\} & l \geq n \end{cases}. \end{aligned}$$

¹That 0 is the smallest number is an axiom of simple arithmetic.

By the collapsing lemma, the sentences in the standard model of arithmetic are still true. One such sentence is $n \neq n + m$ for $m > 0$, because of the meaning of the successor function. In the collapsed model, the sentence $n = n + m$ is true because $n \equiv n + m \pmod{m}$. These two sentences $n = n + m$ and $n \neq n + m$ are contradictory, thus we have an inconsistent model. The truth value of $n = n + m$ is p . This was a straightforward demonstration of an inconsistent model based on simple arithmetic.

3.2 Chunk and Permeate

A way to approach inconsistent systems is to divide them into consistent chunks. This is of non-adjunctive nature because you are not allowed to adjunct statements from different chunks. Brown and Priest [BP04] describe a strategy for still allowing information to permeate between these chunks.

The idea is to only allow a certain subset of sentences in a chunk to permeate into another chunk at a given point in time. To formulate this, let L be the set of sentences of a classical language of first or higher order logic. Let $\vdash$ be the classical deduction relation. For a set Σ of sentences in language L we define $\Sigma^\perp$ to be the closure² of Σ under $\vdash$. For Σ we define a covering to be a partition $\{\Sigma_i \subseteq \Sigma \mid i \in I\}$ for some index set I where $\Sigma = \bigcap_{i \in I} \Sigma_i$ and all Σ_i are consistent. These Σ_i are the chunks that we want to be consistent.

For a covering $C = \{\Sigma_i \mid i \in I\}$ we call ρ a permeability relation on C . This ρ is a map from $I \times I$ to a subset of all sentences of L , so $\rho : I \times I \rightarrow P(L)$ where P is the power set which describes all subsets of L ³. The picture is that we want ρ to map the sentences that are allowed to permeate between the chunks given by the two indices. For $i_0 \in I$, we call the structure $\langle C, \rho, i_0 \rangle$ a C&P structure on Σ .

Next we want to define the C&P consequence relation $\models_{\mathfrak{N}}$. Let $\mathfrak{N} = \langle C, \rho, i_0 \rangle$ be a C&P structure on Σ . For each $i \in I$ and $n \in \mathbb{N}$, we define by recursion:

$$\begin{aligned} \Sigma_i^0 &= \Sigma_i^\perp \\ \Sigma_i^{n+1} &= \left(\Sigma_i^n \cup \bigcup_{j \in I} (\Sigma_j^n \cap \rho(j, i)) \right)^\perp \end{aligned}$$

The intersection $\Sigma_j^n \cap \rho(j, i)$ describes what flows into Σ_i^n from the chunk Σ_j at level n . So Σ_i^{n+1} describes what can be inferred from Σ_i^n together with

²All sentences that can be deduced from Σ .

³This exists according to ZF Set Theory.

what flows into chunk i from the other chunks at level n . At infinity⁴ we get

$$\Sigma_i^\infty = \bigcup_{i < \infty} \Sigma_i^n.$$

We define the C&P consequence $\models_{\mathfrak{N}}$ as follows: $\Sigma \models_{\mathfrak{N}} A$ iff $A \in \Sigma_{i_0}^\infty$ where i_0 marks the designated chunk. This C&P consequence describes all the sentences that can be deduced from the designated chunk i_0 when all the allowed information has been transferred between the chunks. The i_0 does not need to be fixed. Brown and Priest [BP04] mention that this can be context-specific. There can be multiple output chunks, e.g. i_0, i_1 where i_0 is for the conclusion of a microsystem and i_1 for a macrosystem or the i_0, i_1 can only allow certain syntactic forms. An example might be relativistic quantum theory [BP04].

A simple example is a C&P structure that only contains two chunks. For example a source chunk Σ_S and a target chunk Σ_T . The permeability function only allows information to flow from the source to the target. In this case we have that $\Sigma_T^1 = \Sigma_T^\infty$ because all the information can flow in one step.

Brown and Priest apply the Chunk and Permeate method to describe the infinitesimal calculus [BP04] as it was developed by Newton and Leibniz. They also describe Bohr's model of the hydrogen atom [BP15].

3.2.1 Dirac's Delta Function

Here I want to describe the application of C&P by Benham, Mortensen and Priest to the Dirac delta [BMP14]. Dirac introduced the Dirac delta $\delta(x)$ for its application in quantum mechanics [Dir30]. It has three main properties:

- (1) $\int_{-\infty}^{\infty} \delta(x) dx = 1$
- (2) $\delta(x) = 0$ for all $x \neq 0$
- (3) $\int_{-\infty}^{\infty} f(x) \delta(x) dx = f(0)$ for any continuous function f

Thus δ cannot be a normal function over the real numbers. Because of (2), the integral in (1) would have to evaluate to 0 which is not what we want. Dirac himself calls δ a "improper function" [Dir30]. There are numerous mathematically rigorous ways to describe δ , for example via a distribution or a measure.

Dirac makes two arguments for the existence and properties of δ . First that (3) follows from (1) and (2). In his second argument, he shows that the δ

⁴Countably infinite: $\infty = \omega$.

function is a derivative of the Heaviside function Θ^5 by plugging in the derivative of the Heaviside function into (3). From this he shows that (1) and (2) follow. Benham et al. replicate these two arguments with C&P [BMP14].

Let us begin with Dirac's first argument. He claims that the first two statements are true for a function that is non-zero around a small interval around the origin with a sharp peak at zero. Thus in (3) the integral only depends on $f(0)$ without essential error [Dir30]. Benham et al. model this with non-standard analysis [Bel17] where infinitesimally small numbers exist that lie in between zero and the positive real numbers [BMP14]. We set the source theory Σ_S to be nonstandard analysis and the target theory Σ_T standard analysis.

According to Dirac's idea we define $\delta(x)$ to be zero outside a neighborhood of zero

$$(4) \quad \delta(x) = 0 \quad \text{for } x \leq -t < 0 \text{ or } 0 < t \leq x$$

for some positive infinitesimal t . We use (4) instead of (2). To get (1), we define $\delta(x)$ to form a triangle shape around 0 with height $1/t$ and base $2t$ at the origin. Then, the integral (1) evaluates to 1.

Now we aim to derive (3). Let f be any function continuous at zero. In nonstandard analysis that means that

$$(5) \quad \theta \approx 0 \text{ implies } f(\theta) \approx f(0).$$

More formally, for any infinitesimal θ we have

$$(6) \quad f(\theta) = f(0) + \eta(\theta) \quad \text{where } \eta(\theta) \text{ is infinitesimal for all infinitesimal } \theta.$$

For the integration in (3) we only consider a finite interval to make the proof simpler. Let $a \notin (-t, t)$.

$$\begin{aligned}
 \int_{-a}^a f(x) \delta(x) dx &= \int_{-t}^t f(x) \delta(x) dx && \text{because } \delta = 0 \text{ outside of } (-t, t) \\
 &= \int_{-t}^t (f(0) + \eta(x)) \delta(x) dx && x \in (-t, t) \text{ are infinitesimal (6)} \\
 &= f(0) \int_{-t}^t \delta(x) dx + \int_{-t}^t \eta(x) \delta(x) dx \\
 &= f(0) + \int_{-t}^t \eta(x) \delta(x) dx && \text{by (1)} \\
 (7) \quad &= f(0) + \alpha && \text{for some infinitesimal } \alpha
 \end{aligned}$$

⁵Will be defined later, when we use it.

The last step is valid, because for any positive real r and $x \in (-t, t)$ we have $|\eta(x)|\delta(x) < r\delta(x)$. Thus

$$\left| \int_{-t}^t \eta(x)\delta(x) dx \right| \stackrel{\delta \text{ non-negative}}{\leq} \int_{-t}^t |\eta(x)|\delta(x) dx < \int_{-t}^t r\delta(x) dx = r.$$

Therefore, $\alpha = \int_{-t}^t \eta(x)\delta(x) dx$ is infinitesimal since its norm is smaller than any positive rational number.

All these steps are according to the rules in nonstandard analysis. Normally, you would just take the "standard parts" since integrals and derivations in nonstandard analysis are defined by the nearest rational number. The idea from the paper [BMP14] is to permeate the results of (7) into the chunk Σ_T in which all infinitesimals are zero by the Archimedean principle⁶. In chunk Σ_T , the equation (7) becomes (3) and (4) becomes (2), as desired.

Benham et al. note that there is a straight-forward comparison between proofs in nonstandard analysis and the corresponding proofs in C&P. They argue that C&P is not limited to nonstandard analysis. Benham et al. also mention that the equivalence opens the question, if nonstandard analysis or C&P has better explanatory properties.

Dirac's second argument shows the existence of δ as the derivative of the Heaviside function Θ given by:

$$(8) \quad \Theta(x) = \begin{cases} 0 & x < 0 \\ 1 & x > 0 \end{cases}$$

Note that Θ is not defined at $x = 0$. It is discontinuous at $x = 0$ so its derivative is also not defined at $x = 0$, which is what we want for Dirac's delta $\delta(x)$. Dirac wants to show that the derivative Θ' satisfies (3) and that (1) and (2) follow.

$$\begin{aligned} \int_{-a}^a f(x)\Theta'(x) dx &= f(x)\Theta(x)|_{-a}^a - \int_{-a}^a \Theta(x)f'(x) dx && \text{integration by parts} \\ &= f(a) - \left(\int_{-a}^0 0f'(x) dx + \int_0^a 1f'(x) dx \right) && \text{plug in definition (8)} \\ &= f(a) - (0 + f(a) - f(0)) \\ (9) \quad &= f(0) \end{aligned}$$

Thus, the derivative of Θ satisfies (3). By setting $f(x) = 1$, we get (1). And (2) follows from the definition (8) as the derivative of Θ is 0 at points $x \neq 0$.

The problematic part in his argument is that integration by parts requires Θ' and f' to exist at all points in $(-a, a)$, which is not the case for definition (8)

⁶The Archimedean principle states: for all positive numbers $x < y$ there exists a natural number $n \in \mathbb{N}$ such that $nx \geq y$.

we used in the second step. Thus there is a contradiction in Dirac's second argument [BMP14].

In C&P there is a simple argument. In the source theory Σ_S we take Θ to be a differentiable function. Let h be some positive real number. We define

$$(10) \quad \Theta(x) := \begin{cases} 0 & x \leq -h \\ 2(1 + x/h) & -h \leq x \leq h \\ 1 & x \geq h \end{cases}.$$

This function Θ goes in a "straight line" from 0 to 1.

The first step in (9), the integration by parts, we do with (10). The result is then sent to the target theory Σ_T where $h = 0$ such that Θ is the step function. Thus we get (3).

The above argument to model Dirac's second argument is based only on standard analysis and does not have a nonstandard source theory. Benham et al. point out that for different arguments, the source theories are generally also different. In Dirac's first argument, a nonstandard analysis source theory was used, whereas for the second argument we used a standard analysis theory. Benham et al. conclude by stating that C&P is a straightforward model for Dirac's idea of the delta function.

Chapter 4

Philosophy of Paraconsistency

This last chapter is about philosophical aspects of paraconsistency. First, we explore where paraconsistent reasoning is applied and which systems benefit from it. Then, how mathematicians use logic in their work. The last part is about the existence of contradiction and the comparison of bivalence and dialetheism.

4.1 Motivations for Paraconsistency

The main objective of paraconsistent logic according to Brown [Bro02] is to prevent triviality in a system where a contradiction can occur or where the premisses will lead to a contradiction. Yet the specific motivations differ from person to person and between different application. One form of application is to model a system where we might not have enough data to evaluate the exact truth of sentences. Another application is to draw conclusions from inconsistent premisses. Dialetheists belief that the world is inconsistent, thus the true logic should be paraconsistent. I see paraconsistency as a tool to handle inconsistencies in a passive approach, without trivializing a potentially interesting theory.

Evaluation, of whether or not a given set of premisses is consistent, is a difficult task. First we need to determine all the sentences that syntactically follow from these premisses. Then, we need to find a model where all these sentences are also true. There is no systematic way to find such a model as this problem is not decidable [Bro02]. Also there might be reasons why an inconsistent system is preferable. For example if it is exceptionally simple like Bohr's hydrogen model or the naive set theory.

There are cases where a paraconsistent logic seems like a good fit. For instance in epistemic logic, which is concerned with logical approaches to knowledge. To formalize inconsistent beliefs, we need a paraconsistent logic,

otherwise we get a trivial theory. When we notice conflicting beliefs we might change one or many of them to achieve consistency. But it is also possible not to change your contradictory beliefs. This does not inherently hinder our reasoning capabilities. Thus, I think, we are reasoning in a paraconsistent fashion. To model such a belief system we need a paraconsistent logic.

Another example is deontic logics, which are concerned with permissions and obligations. This appears in moral questions or laws. Assume you have a set of laws and somehow these lead to a contradiction. Then we would want a judge to reason in a paraconsistent way, to prevent everything from being legal and illegal at the same time. Perhaps the paraconsistent nature of law is why we need so many lawyers. Gödel himself made remarks that the US constitution is inconsistent [GP13].

These examples alone are motivation enough for me to claim that paraconsistent logics are worth exploring. Note that the above examples have a strong connection to humans. It might be the case that we as human beings and our thoughts are not consistent themselves or too complex to be fully understood. Aristotle argued that humans are defined by our ability to have rational thoughts and named humans the rational animal [AM21]. Perhaps we are not as rational as we like to be and the idea of rigorously applying classical logic to human endeavours was doomed from the beginning.

Brown notes that there are also physical theories which follow a paraconsistent logic. One such example is the description of the hydrogen atom by Bohr [BP15]. In this model the nucleus of the atom is very small at the center of the atom and electrons revolve around it in different circular orbitals. Bohr's model is not consistent¹, because we cannot know the exact position and momentum of an electron at the same time². Nevertheless, it helped to describe a phenomenon people at the time were trying to understand. Another inconsistency in physics is between general relativity and quantum mechanics [Bro02]. Yet these theories still predict what we can empirically measure. Thus they have great value, even though they cannot be described with classical logic without being trivialized.

As we saw with the Dirac delta in 3.2.1, in physics empirical usefulness of theories can be more important than their consistency. Thus classical logic does not truly resemble the ideas or thinking processes of these physicists. And acceptance of these inconsistencies makes their logic paraconsistent. This does not necessarily mean that the physical world has to be inconsistent, just that the frontier of research can work with paraconsistent theories.

A better understanding of paraconsistency could help people working at the

¹Based on our current understanding.

²By the Heisenberg uncertainty principle about position and momentum or by the wave particle duality amongst other principles.

frontier of any endeavor to be more secure about what they know. And what role the inconsistent parts play. Maybe there are rich theories we can only discover if we allow a great deal of inconsistency.

Another motivation for paraconsistency are systems that are per construction inconsistent systems. A real-world example would be a data collection where some of the data contradict each other. To use this data for prediction it can be advantageous to use a paraconsistent logic [FB19]. Humans can also create deliberately inconsistent art.

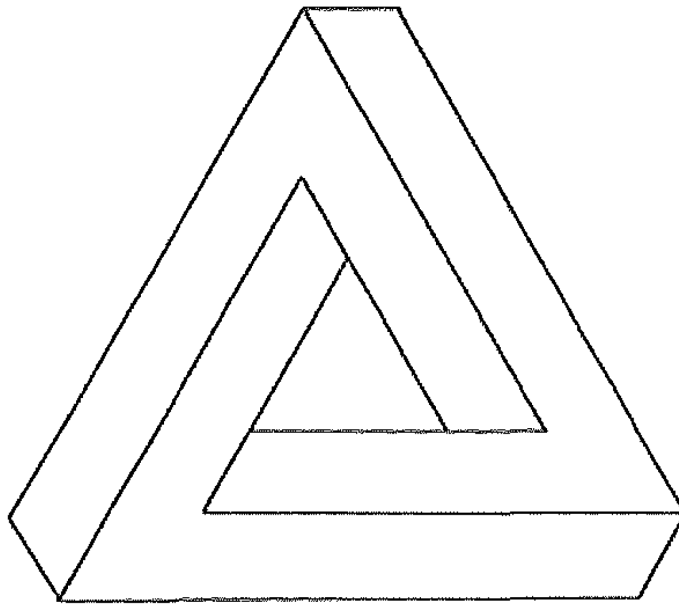


Figure 4.1: Penrose triangle from [Mor10] on page 117.

4.2 What Logic is used by Mathematicians?

I intentionally left out mathematics in the treatment above. Because unlike the examples already discussed, mathematics does not need to be based on anything in reality. In some ways, mathematics is just logic applied to itself. What logic is used by mathematics might be too difficult a question to answer. There are many mathematicians and they certainly do not agree³ on everything [Hor19]. But we can investigate how close the way of thinking of mathematicians is to classical logic.

In my reflection about the formalization of paraconsistent logics in section 2.4

³From a philosophical standpoint the four main schools are: logicism, intuitionism, formalism and predicativism.

I already mentioned that the inference rules of classical logic are somewhat of a necessity for mathematics. They build the basic logical steps that are used in mathematical proofs. Because mathematics can be described as a formal system there are very successful attempts at formalizing mathematics with classical logic. This is done with first-order logic and set theory. Most of mathematics can be described in this way [Bag20].

But here, the first problem arises concerning how I think about mathematics. In my studies as a mathematician I was taught naive set theory in the first semester. With the caveat that there exist different set theories that have greater robustness than naive set theory and a quick demonstration of Russell's paradox. In most proofs I read or wrote, my picture of a set can be described by naive set theory. The reason for this is that I only know a handful of the axioms of Zermelo-Frankel set theory and I feel that in a lot of proofs it does not matter what set theory you think about.

Another such problematic example is Leibniz' calculus [BP04] where we work with infinitesimal numbers that are treated sometimes like zero and sometimes bigger than zero. This is an old way of thinking about integrals and derivatives that is now superseded by a more formal theory based on Riemann and Cauchy. But the Leibniz' way of thinking can help you to build an intuition about integrals. Especially if you only consider two or three dimensions. So it is still commonly used, especially in more application-oriented circumstances such as physics. Personally I think this has great explanatory value.

Cantor and Leibniz, who developed these two theories, were influential mathematicians and some of their work is still taught and used today. Brown states that classical logic cannot be used to model their theories because they would become trivial [Bro02]. The development based on inconsistent theories is still valuable for mathematics. At least for explanatory purposes, simplicity or understanding.

Mathematicians seem to prefer consistency. The development of modern analysis to consistently describe integration and derivation is beautiful and interesting in its own right. A similar development happened to Dirac's delta, which is mathematically clearly not a function. This led to the development of distributions by Laurent Schwartz [BMP14]. This is a rich area of mathematics, where the Dirac delta exists in a consistent way.

This opens some questions. Clearly, the Dirac delta was useful before it was consistently formalized in mathematics. But was it also valid or true before the formalization? Schwartz developed a theory that could describe Dirac's delta, but what if no one managed to derive such a theory? Can a mathematical theory be valuable if it is clearly inconsistent? Cantor and Leibniz knew about the inconsistencies in their theory, yet still argued that

they have value to mathematics. Here is a quote by Wittgenstein that Priest seems to love⁴:

Indeed, even at this stage, I predict a time when there will be mathematical investigations of calculi containing contradictions, and people will actually be proud of having emancipated themselves from consistency. (Wittgenstein, 1930)

I think mathematicians think paraconsistently as it is just another form of epistemology described in the section above. However, their goal seems to be to create a consistent mathematics. Such a consistent mathematics has huge potential, as we can see in computers or other consistent scientific theories. Nevertheless it is worth exploring inconsistent theories. Perhaps we can find some rich insight or beauty in a mathematical description of inconsistent systems, which ought to have a paraconsistent logic.

4.3 Do Inconsistencies Exist?

After all this we might want to ask if inconsistencies actually exists. Bivalence and dialetheism are two opposing stances in formal logic. Bivalence is the classical thinking that a sentence is either true or false, which is strongly related to the law of the excluded middle $A \vee \neg A$ or the classical tautology called law of non-contradiction $\neg(A \wedge \neg A)$. This idea goes back to ancient Greek philosophy. Most prominently, Aristotle states in his *Metaphysics* that the law of non-contradiction is "the most certain of all principles" [Hor18]. Since then there have not been a lot of arguments in favor of the law of non-contradiction. Its validity has been taken for granted [PBW18].

The argument for dialetheism and true contradictions starts with the claim that a priori there is no reason for a sentence to be exclusively true or false. This is merely a historic development in western philosophy. The idea of dialetheism is arguably as old as its counterpart. For example, Heraclitus' views were considered by Plato and Aristotle to be in violation of the law of non-contradiction [Gra21].

Far away from Greece, in ancient India, a form of logic was developed called *Catuskoṭi*. In *Catuskoṭi* a statement can be true, false, neither true or false or both true and false. In some cases there was even a fifth possibility: *non of the others*. Also in Daoism in China, contradictory comments can be found [PBW18]. Zen is a philosophy which inherits from Daoism and Buddhism. It incorporates expressions that should not be interpreted with an Aristotelian either-or approach [Nag20]. An example:

The river does not flow but the bridge does.

⁴He uses this quote as an epigraph in some of this work e.g. [Pri00][Pri79].

It seems that it is also a cultural question of how truth values should be handled. Our opinion might be more influenced by the society we grew up in than by an a priori fact regarding how truth has to behave.

In modern times, Hegel has been seen as one of the first critics of the law of noncontradiction. He saw himself as a successor to Heraclitus. See [Hor18] for in depth information about the whole history. The recent development in paraconsistency and dialetheism came from South America with Florencio Asenjo and Newton da Costa. It has spread over the whole world and also into different scientific studies.

Dialetheists do not claim that every contradiction is true. They know the effectiveness of consistent systems and want to preserve it with paraconsistent logics. The motivation is that they believe that inconsistencies actually exist in this world and that we have to accept them. Examples like the Liar paradox or inconsistent picture show that the human consciousness is able to go beyond consistency [RY19].

To me it seems that the existence of paradoxes such as the liar paradox or inconsistent drawings show that humans are capable of creating inconsistencies. And it does not come with great effort. This is enough for me to say that there are systems that need to be approached paraconsistently. Another example are inconsistent datasets. Imagine temperature measurement at different times. But at one specific time there are two measurements with different temperatures.

What about the the world we experience? Is it truly consistent? This is a tricky question because you can twist the definitions around to fit your own perspective. For example, there are numerous ways of dealing with the liar paradox [BGR20]. Paraconsistency is just one of many non-classical approaches. There are also classical solutions such as Tarski's hierarchy of languages[BGR20]. There is no definitive consensus.

Arguing against dialetheism is not difficult. We just have to say that if you have inconsistent premisses, they cannot all be true. Thus, we would need to find a new set of premisses which are consistent. I touched on this in the sections about the motivations for paraconsistency.

'From a contradiction everything would follow.' The reply to that is: Well then, don't draw any conclusions from a contradiction.
(Wittgenstein, 1939)

Chapter 5

Conclusion

A contradiction in a classical system makes every sentence we can formulate in such a system trivial by the principle of explosion. The primary goal of paraconsistent logic is to prevent the trivialisation of a theory because of a contradiction. There are multiple approaches to formalize paraconsistent system. The logic of paradoxes allows sentences to be both true and false, but then the inference rules do not stay valid in general. In Chunk and Permeate, we split a theory into consistent Chunks and let information permeate between these chunks. C&P can be used to model inconsistent theories, while the different chunks are still consistent.

The motivations to apply paraconsistent logic is given by inconsistent systems developed by humans. Some examples are inconsistent laws, contradictory personal beliefs, Bohr's model of the hydrogen atom or naive set theory. These theories cannot be described with classical logic without trivializing them. Perhaps human thinking allows inconsistencies to exist.

In mathematics, there seems to be a strife for consistency. Seen in the development of axiomatic set theories, modern analysis or the theory of distributions to describe Dirac's delta. Mathematicians nowadays might be too focused on the ideal of consistency. Leibniz and Cantor knew of the problematic parts in their theories yet still defended them. The further exploration of paraconsistent logics could bring a paradigm shift, which allows parting away from the idealisation of consistency.

Appendix A

Appendix

A.1 Deduction via Tarski Truth Scheme

From the sentence $T(1) \leftrightarrow (T(1) \rightarrow A)$ we get the two sentences

$$\begin{aligned} T(1) &\rightarrow (T(1) \rightarrow A) \\ T(1) &\leftarrow (T(1) \rightarrow A). \end{aligned}$$

Note that the second one is the same as $(T(1) \rightarrow A) \rightarrow T(1)$. From these premisses we can deduce A with the inference rules modus ponens (MP) and absorptoin (abs.):

$$\frac{\frac{(T(1) \rightarrow A) \rightarrow T(1)}{T(1)} \quad \frac{T(1) \rightarrow (T(1) \rightarrow A)}{T(1) \rightarrow A} \text{ (MP)}}{A} \text{ (abs.)} \quad \frac{T(1) \rightarrow (T(1) \rightarrow A)}{T(1) \rightarrow A} \text{ (abs.)}$$

Important to note is that at the top of this tree like structure there are only the two sentences described above. Which we assume to be true.

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